

Proposal Selection in MCMC

From Optimal Scaling to Adaptive Methods

统计计算 第六组

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Outline

Outline

Motivation

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Random-Walk Metropolis Algorithm

算法 Random walk Metropolis

1: Starting with

$$X^{(0)} := (X_1^{(0)}, \dots, X_p^{(0)})$$

and using a symmetric distribution g , iterate for $t = 1, 2, \dots$

2: Draw $\epsilon \sim g$ and set

$$X = X^{(t-1)} + \epsilon.$$

3: Compute

$$\alpha(X | X^{(t-1)}) = \min \left\{ 1, \frac{f(X)}{f(X^{(t-1)})} \right\}. \quad (1)$$

4: With probability $\alpha(X | X^{(t-1)})$ set $X^{(t)} = X$; otherwise set $X^{(t)} = X^{(t-1)}$.

- As $q(X | X^{(t-1)}) = g(X - X^{(t-1)}) = g(X^{(t-1)} - X) = q(X^{(t-1)} | X)$ using the symmetry of g .

Proposal Distribution

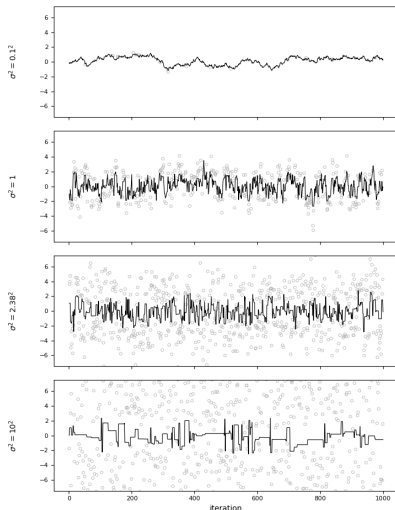
Goal: Small correlation of $X^{(t-1)}$ and $X^{(t)}$

- Correlation between the current state $X^{(t-1)}$ and the newly proposed value $X \sim q(\cdot | X^{(t-1)})$.
- Correlation introduced by retaining the current value $X^{(t)} = X^{(t-1)}$ when the newly generated value X is rejected.

Trade-off: Fast changes vs. high acceptance

- **Small proposal variance:** the probability of acceptance is *high*, but the resulting Markov chain is *highly correlated*, since $X^{(t)}$ changes only very slowly.
- **Large proposal variance:** the chain can potentially move very *fast*, but the probability of acceptance is rather *low*.

Intro Experiment 1: Scale Changes the Chain



- Target density: $N(0, 1)$.
- Random-walk proposal:

$$X^* = X^{(t-1)} + \epsilon, \quad \epsilon \sim N(0, \sigma^2).$$

Open grey discs represent rejected values.

Different Choices of σ^2 in Intro Experiment 1

Proposal variance	Autocorrelation mean	Autocorrelation 95% CI	Acceptance mean	Acceptance 95% CI
$\sigma^2 = 0.1^2$	0.9895	(0.9884, 0.9907)	0.9703	(0.9684, 0.9723)
$\sigma^2 = 1$	0.7686	(0.7628, 0.7743)	0.7049	(0.7021, 0.7077)
$\sigma^2 = 2.38^2$	0.6262	(0.6196, 0.6329)	0.4432	(0.4398, 0.4466)
$\sigma^2 = 10^2$	0.8379	(0.8323, 0.8435)	0.1254	(0.1230, 0.1278)

* Averaged over 100 runs of the algorithm.

Interpretation

- Choosing σ^2 too small yields a very high probability of acceptance, however at the price of a chain that is hardly moving.
- Choosing σ^2 too large allows the chain to make large jumps, however most of the proposed values are rejected.
- The goal is not the highest acceptance rate, but lower autocorrelation and higher sampling efficiency.

Efficient Metropolis Jumping Rules (Gelman, Roberts & Gilks, 1996)

Main question

What are the most efficient symmetric jumping kernels for simulating a normal target distribution using the Metropolis algorithm?

- This paper focuses on proposal **scale**.
- For a d -dimensional spherical multivariate normal problem, the optimal symmetric jumping kernel has scale approximately $\frac{2.4}{\sqrt{d}}$ times the scale of the target distribution¹.
- The associated optimal acceptance rate is about 44% for $d = 1$ and declines to 23% as $d \rightarrow \infty$.
- The efficiency of the Metropolis algorithm compared with independent samples from the target distribution is approximately $0.3/d$.

¹Gelman, Roberts, and Gilks 1996.

Univariate Examples: Two Efficiency Measures

Definitions

- ① For a one-dimensional chain, the limiting scaled sample variance from the Markov chain output is $V_{\bar{\theta}} = \lim_{N \rightarrow \infty} N \text{Var}(\bar{\theta})$, where $\bar{\theta} = \frac{1}{N} \sum_{t=1}^N \theta^{(t)}$ is the sample mean. The efficiency of estimating the mean is

$$\text{eff}_{\theta} = \frac{\tau^2}{V_{\bar{\theta}}} = [1 + 2(\rho_1 + \rho_2 + \cdots)]^{-1}.$$

Approximated by truncating it at a finite lag K : $V_{\bar{\theta}} \approx \gamma_0 + 2 \sum_{t=1}^K \gamma_t$.

- ② A well known inequality^a for eff_{θ} , $\text{eff}_{\bar{\theta}} = \left\{ \sum_{i=1}^{\infty} a(i) \frac{1+\lambda_i}{1-\lambda_i} \right\}^{-1} \geq \frac{1-\lambda_2}{1+\lambda_2}$. So define

$$\text{eff}_{\text{eig}} = \frac{1 - \lambda_2}{1 + \lambda_2}.$$

^aBesag and Green 1993.

- ρ_t is the lag- t autocorrelation and γ_t is the lag- t autocovariance.
- λ_2 is the second-largest eigenvalue of the transition kernel.
- Lower autocorrelation leads to a smaller asymptotic variance and hence higher efficiency.

Simulation 1: Setup

Target and proposal

$$\pi(\theta) = \mathcal{N}(0, 1),$$

$$\theta^* = \theta^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2).$$

- Use a discrete approximation to π : replace the sample space of θ by a discrete array of 100 points spread evenly over $[-6, 6]$.
- The normal density is computed on these points and then renormalized.

Transition matrix

Let

$$Q_{ij} = P(\theta^{\text{prop}} \in C_j \mid \theta_i) \approx q(\theta_j \mid \theta_i) \Delta \theta$$

be the discretized proposal probability, where C_j is the grid cell centered at θ_j .

For $i \neq j$, the Metropolis transition probability is

$$P_{ij} = Q_{ij} \alpha(\theta_i, \theta_j), \quad i \neq j.$$

The diagonal is filled by the remaining probability

$$P_{ii} = 1 - \sum_{j \neq i} P_{ij}.$$

Simulation 1: Discretized Proposal Probabilities

```
def jump_probabilities(values, proposal_sd):  
    # values = numpy.linspace(-6, 6, 100)  
    grid_step = values[1] - values[0]  
    proposal_density = norm.pdf(  
        # Matrix: proposal[j] - current[i]  
        values[None, :] - values[:, None],  
        loc=0,  
        scale=proposal_sd  
    )  
    return proposal_density * grid_step
```

- $\text{values}[\text{None}, :]$: $1 \times n$, all possible proposal candidate values.
- $\text{values}[:, \text{None}]$: $n \times 1$, all possible current values.
- Python broadcasting gives the full proposal matrix.

$$Q_{ij} = P(\theta^{\text{prop}} \in C_j \mid \theta_i) \approx q(\theta_j \mid \theta_i) \Delta\theta.$$

Multiplying by $\Delta\theta$ converts the continuous proposal density into an approximate proposal probability.

Simulation 1: Acceptance Rate

Acceptance Rate

For the normal kernel and normal target distribution, the acceptance rate can be determined analytically

$$p_{\text{jump}} = \frac{2}{\pi} \arctan\left(\frac{2}{\sigma}\right)$$

Because the proposal is symmetric,

$$p_{\text{jump}} = 2P\{\pi(\theta') \geq \pi(\theta)\} = 2P\{|\theta'| \leq |\theta|\}.$$

Now

$$|\theta + \epsilon| \leq |\theta| \iff \epsilon(2\theta + \epsilon) \leq 0.$$

Let $U = \epsilon/\sigma \sim N(0, 1)$. Then (θ, U) is rotationally symmetric, and the boundary lines are

$$U = 0, \quad U = -\frac{2}{\sigma}\theta.$$

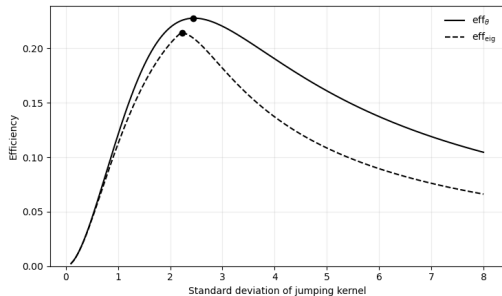
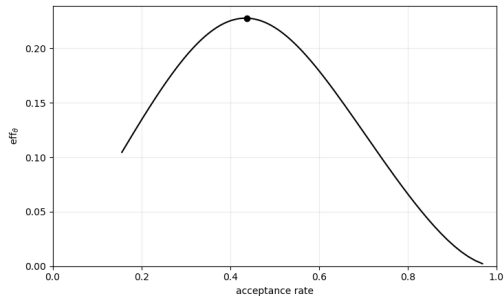
Their angular width is $\arctan(2/\sigma)$, so

$$P\{|\theta'| \leq |\theta|\} = \frac{1}{\pi} \arctan\left(\frac{2}{\sigma}\right),$$

and therefore

$$p_{\text{jump}} = \frac{2}{\pi} \arctan\left(\frac{2}{\sigma}\right)$$

Simulation 1: Normal Kernel Results

Efficiency as functions of σ  eff_{θ} as a function of acceptance rate

best σ	eff_{θ}	eff_{eig}	acceptance rate
2.4352	0.2277	0.2097	0.4377
2.2365	0.2259	0.2145	0.4645

Simulation 1: Other Kernels

Uniform Kernel

A symmetric uniform kernel with width $\sqrt{12}\sigma$ and thus a variance of σ^2 .

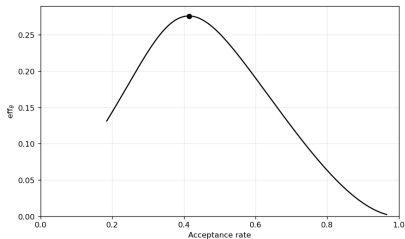
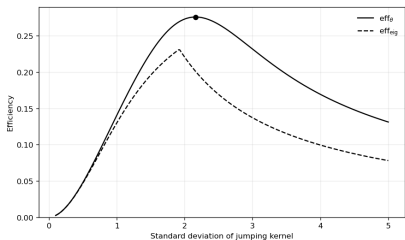
Bimodal Kernel

- A mixture of two normal densities with $J \sim \frac{1}{2}N\left(-\frac{\sigma}{\sqrt{2}}, \frac{\sigma^2}{2}\right) + \frac{1}{2}N\left(\frac{\sigma}{\sqrt{2}}, \frac{\sigma^2}{2}\right)$.
- The bimodal kernel is symmetric and has variance σ^2 .
- The idea was to reduce the number of very small jumps, which may lower the efficiency of the Metropolis algorithm.

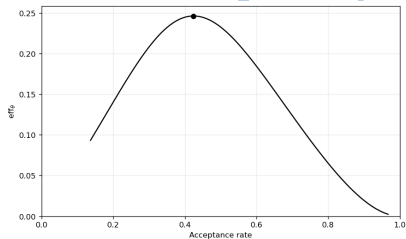
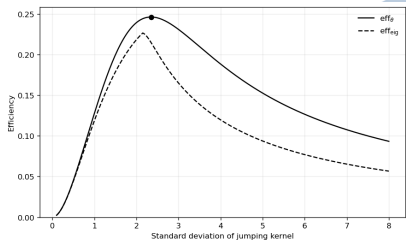
Kernel	best σ	eff_θ	eff_{eig}	acceptance rate
Uniform	2.1583	0.2760	0.2014	0.4141
Mixture normal	2.3477	0.2462	0.2147	0.4238

The results are very close to those obtained with the normal kernel.

Simulation 1: Other Kernels Results



Uniform kernel



Bimodal kernel

Theorem 3.1: Multivariate Target Distributions

Setup

- Consider a product target $\pi(\theta) = \prod_{i=1}^d f(\theta_i)$ *without* the assumption of normality.
- Use a **multivariate normal** proposal kernel with variance-covariance matrix $\sigma_d^2 I_d = \frac{\phi^2}{d} I_d$.

A Speeded-Up & Continuous-Time Version

$$Y_t^d = \theta_1^{[td]}$$

- Only consider the first component of θ .
- Remains constant for a time interval $1/d$ before making a jump.
- Y^d is in general not a Markov process unless $d \rightarrow \infty$.

Fisher's Information Measure for f

$$\begin{aligned}
 F &= \int_{-\infty}^{\infty} \frac{(f'(x))^2}{f(x)} dx \\
 &= \mathbb{E} \left[\left(\frac{d}{dx} \log f(X) \right)^2 \right]
 \end{aligned}$$

- For standard normal f , we have $F = 1$.

Theorem 3.1: Multivariate Target Distributions

Theorem 3.1

As $d \rightarrow \infty$, assuming the starting value of $\theta_1, \dots$ are all distributed independently according to f , the process Y^d converges weakly to the limiting Langevin diffusion

$$dY_t = \frac{f'(Y_t)}{2f(Y_t)} h(\phi) dt + \sqrt{h(\phi)} dB_t,$$

where

$$h(\phi) = 2\phi^2 \Phi\left(-\frac{\phi\sqrt{F}}{2}\right)$$

- The limiting value of p_{jump} for this sequence of problems is $h(\phi)/\phi^2$.
- Optimizing eff_θ is equivalent to optimizing $h(\phi)$ (is called a *speed* measure²).
- The simulation shows that the conclusions above are also approximately true for dimension as low as 6.

²Roberts, Gelman, and Gilks 1997.

Theorem 3.1: Optimizing $h(\phi) = 2\phi^2\Phi\left(-\frac{\phi\sqrt{F}}{2}\right)$

Optimal scale

- ① **Reparameterization** Let $z = \frac{\phi\sqrt{F}}{2}$, then maximizing $h(\phi)$ is equivalent to maximizing

$$z^2\Phi(-z)$$

- ② **Differentiate and Set to Zero**

$$2\Phi(-z) = z\varphi(z)$$

where φ is the standard normal density.

- ③ **Solution** Solving the equation gives $z \approx 1.19$.

$$\hat{\phi} \approx \frac{2.38}{\sqrt{F}}$$

Optimal acceptance probability

The limiting acceptance probability is

$$p_{\text{jump}}(\phi) = \frac{h(\phi)}{\phi^2} = 2\Phi\left(-\frac{\phi\sqrt{F}}{2}\right)$$

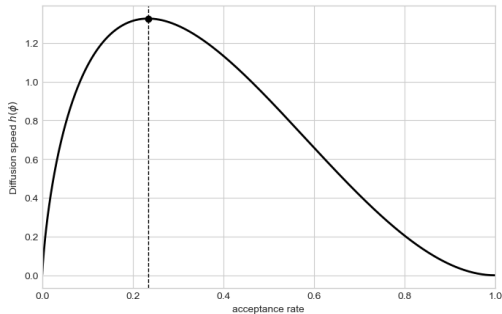
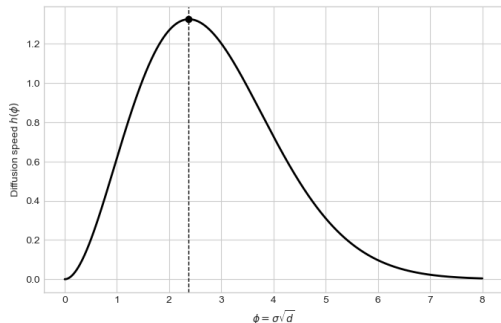
At the optimal scale,

$$p_{\text{jump}}(\hat{\phi}) = 2\Phi(-1.19) \approx 0.234$$

Standard Normal Target

$$F = 1 \Rightarrow \sigma_d \approx \frac{2.38}{\sqrt{d}}$$

Theorem 3.1: Figures



- Efficiency (or diffusion speed $h(\phi)$), is maximised by setting $\phi \approx 2.38$.
- The optimal limiting acceptance rate is approximately 0.234.
- Acceptance rates between 0.15 and 0.40 yield at least 80% of the maximum efficiency.

Simulation 2: Finite Dimensions Set Up

- Target distribution:

$$\pi(\theta) = N_d(0, I_d)$$

- Asymptotically optimal proposal standard deviation:

$$\sigma_d \approx \frac{2.38}{\sqrt{d}}$$

- The chain is initialized from stationarity

$$\theta^{(0)} \sim N_d(0, I_d)$$

so that no additional burn-in correction is needed for estimating the stationary efficiency.

- 21 values are taken between $\frac{2}{\sqrt{d}}$ and $\frac{3}{\sqrt{d}}$.
- Use 10^6 iterations.



Simulation 2: Efficiency

Estimating Efficiency by Batch means

Let the chain length be $N = mb$, where m is the number of batches and b is the batch size. For coordinate j , define the mean of batch r

$$\bar{\theta}_{j,r}^{(b)} = \frac{1}{b} \sum_{t=(r-1)b+1}^{rb} \theta_j^{(t)}, \quad r = 1, \dots, m.$$

If b is sufficiently large,

$$\text{Var}(\bar{\theta}_{j,r}^{(b)}) \approx \frac{\bar{V}_j}{b} \iff \bar{V}_j \approx b \cdot \text{Var}(\bar{\theta}_{j,r}^{(b)})$$

$$\hat{V}_j = b \cdot \frac{1}{m-1} \sum_{r=1}^m \left(\bar{\theta}_{j,r}^{(b)} - \bar{\bar{\theta}}_j^{(b)} \right)^2$$

where $\bar{\bar{\theta}}_j^{(b)} = \frac{1}{m} \sum_{r=1}^m \bar{\theta}_{j,r}^{(b)}$.

Benchmark: $\text{eff}_{\max} \approx 0.331/d$

For a standard normal target, $h(\phi) = 2\phi^2 \Phi\left(-\frac{\phi}{2}\right)$.

At the optimal scale,

$$\phi \approx 2.38, p_{\text{jump}} \approx 0.234 \Rightarrow h_{\max} \approx 1.325$$

Since the original chain is slowed down by d ,

$$\text{eff}_{\max} \approx \frac{h_{\max}}{4d} \approx \frac{0.331}{d}$$

- For a spherical target $\widehat{\text{eff}} = \frac{1}{d} \sum_{j=1}^d \widehat{\text{eff}}_j$,
- This reduces the impact of random variation in determining the optimal σ_d .
- With batch size 100, first-order autocorrelations in batch means were always less than 0.1.

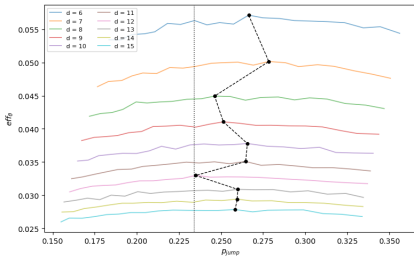
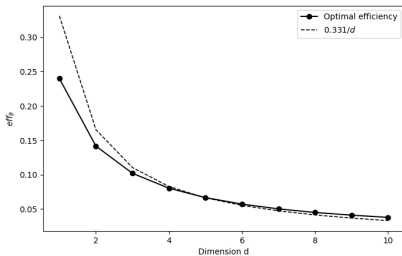
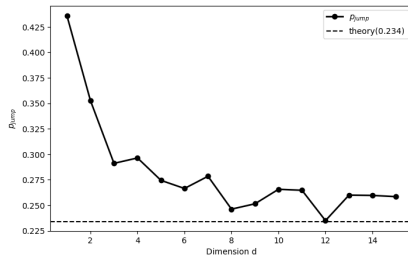
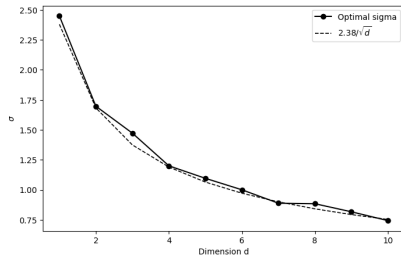
Simulation 2: Results

d	Optimal σ_d	eff_θ	p_{jump}	$2.38/\sqrt{d}$	$0.331/d$
1	2.45	0.240	0.436	2.38	0.331
2	1.70	0.142	0.353	1.68	0.166
3	1.47	0.102	0.291	1.37	0.110
4	1.20	0.080	0.296	1.19	0.083
5	1.10	0.067	0.274	1.06	0.066
6	1.00	0.057	0.266	0.97	0.055
7	0.89	0.050	0.278	0.90	0.047
8	0.88	0.045	0.246	0.84	0.041
9	0.82	0.041	0.251	0.79	0.037
10	0.74	0.038	0.266	0.75	0.033

The value of σ_d giving largest average efficiency eff_θ for each d together with the corresponding empirical proportion of accepted jumps p_{jump}

- The optimal σ_d is close to $2.38/\sqrt{d}$ while p_{jump} moves toward the 0.234 rule.
- From $d = 5$ onward, eff_θ is already close to $0.331/d$.

Simulation 2: Results



Intro Experiment 2: Shape can also make difference

Example 9.2 (Bayesian probit model). In a medical study on infections resulting from birth by Caesarean section (taken from Fahrmeir and Tutz, 2001)  three influence factors have been studied: an indicator of whether the Caesarian was planned or not (z_{i1}), an indicator of whether additional risk factors were present at the time of birth (z_{i2}), and an indicator of whether antibiotics were given as a prophylaxis (z_{i3}). The response Y_i is the number of infections that were observed amongst n_i patients having the same influence factors (covariates). The data is given in Table [9.1](#).

Table 9.1: Data used in example 9.2

Number of births with infection		planned	risk factors	antibiotics
y_i	n_i	z_{i1}	z_{i2}	z_{i3}
11	98	1	1	1
1	18	0	1	1
0	2	0	0	1
23	26	1	1	0
28	58	0	1	0
0	9	1	0	0
8	40	0	0	0

The posterior density of β which we want to sample from

$$\prod_{i=1}^N \Phi(z'_i \beta)^{y_i} (1 - \Phi(z'_i \beta))^{n_i - y_i} \exp \left(-\frac{\lambda}{2} \sum_{j=0}^3 \beta_j^2 \right)$$

- A suitable prior distribution for the parameter of interest is $\beta \sim N(0, \mathbb{I}/\lambda)$.
- Guided by the GLM asymptotic covariance, another choice is $\Sigma \propto (Z'DZ)^{-1}$.

Proposal	Acc.	β_0	β_1	β_2	β_3
$\Sigma = 0.08I$	0.1376	0.9467	0.9505	0.9525	0.9524
$\Sigma \propto (Z'DZ)^{-1}$	0.2022	0.8615	0.8745	0.8649	0.8751

Autocorrelation $\rho(\beta_j^{(t-1)}, \beta_j^{(t)})$ between subsequent samples.

- The autocorrelation is reduced and the acceptance rate increases to 20%.
- The determinant was chosen to be the same, so the improvement is due to a difference in the structure of the covariance.

From Scale to Shape: Related Developments Before Adaptive Metropolis

Year	Work	Contribution
1998	Gilks, Roberts & Sahu <i>Adaptive MCMC</i>	Uses regeneration / atom-based ideas to adapt proposal mechanisms while preserving correct ergodic properties. Shows that adaptation is possible, but must be carefully controlled.
1999	Haario, Saksman & Tamminen <i>Adaptive Proposal Distribution</i>	Introduces an adaptive proposal strategy using recent history of the chain. This moves beyond tuning only the scalar step size and begins to learn proposal covariance structure.
1999	Sahu & Zhigljavsky <i>Self-regenerative adaptive MCMC</i>	Develops a self-regenerative adaptive framework, using auxiliary structure to allow proposal tuning without losing the target distribution.
1999–2000	Early covariance-based and componentwise tuning ideas	A common practical theme emerges: different coordinates and directions need different proposal scales. Acceptance-rate tuning alone is not enough when the target has strong anisotropy or correlation.
2000	Breyer & Roberts <i>Optimal scaling for Markov random fields</i>	Shows that scaling rules can extend beyond independent product targets to locally dependent structures, but only under weak-dependence or no phase-transition conditions.

Adaptive Metropolis Algorithm

算法 Adaptive Metropolis³

- 1: Starting with an initial point $X_0 \in S \subset \mathbb{R}^d$, an initial positive definite matrix C_0 , a threshold t_0 , and a small constant $\varepsilon > 0$.
- 2: For $t = 1, 2, \dots$, define the proposal covariance matrix

$$C_t = \begin{cases} C_0, & t \leq t_0, \\ s_d \text{cov}(X_0, \dots, X_{t-1}) + s_d \varepsilon I_d, & t > t_0, \end{cases} \quad s_d = \frac{(2.4)^2}{d}$$

- 3: Generate a candidate point $Y \sim N(X_{t-1}, C_t)$.
- 4: Compute the acceptance probability $\alpha(X_{t-1}, Y) = \min \left\{ 1, \frac{\pi(Y)}{\pi(X_{t-1})} \right\}$.
- 5: With probability $\alpha(X_{t-1}, Y)$ set $X_t = Y$; otherwise set $X_t = X_{t-1}$.

- The AM process is **non-Markovian**, since C_t depends on the history $X_0, \dots, X_{t-1}$.
- The size of t_0 reflects our trust in C_0 . The bigger it is chosen the more slowly the effect of the adaptation is felt.
- $\varepsilon > 0$ is a small constant added to prevent the covariance matrix from becoming singular.

³Haario, Saksman, and Tamminen 2001.

Update of the C_t with No Extra Computational Cost

For $x_0, \dots, x_k \in \mathbb{R}^d$, $\text{cov}(x_0, \dots, x_k) = \frac{1}{k} \left(\sum_{i=0}^k x_i x_i^T - (k+1) \bar{x}_k \bar{x}_k^T \right)$.

For $t > t_0 + 1$,

$$C_t = \frac{s_d}{t-1} \left(\sum_{i=0}^{t-1} X_i X_i^T - t \bar{X}_{t-1} \bar{X}_{t-1}^T \right) + s_d \varepsilon I_d \implies \sum_{i=0}^{t-1} X_i X_i^T = \frac{t-1}{s_d} (C_t - s_d \varepsilon I_d) + t \bar{X}_{t-1} \bar{X}_{t-1}^T$$

Using

$$\sum_{i=0}^t X_i X_i^T = \sum_{i=0}^{t-1} X_i X_i^T + X_t X_t^T,$$

we obtain

$$C_{t+1} = \frac{t-1}{t} C_t + \frac{s_d}{t} [t \bar{X}_{t-1} \bar{X}_{t-1}^T - (t+1) \bar{X}_t \bar{X}_t^T + X_t X_t^T + \varepsilon I_d].$$

$$\bar{X}_t = \frac{t \bar{X}_{t-1} + X_t}{t+1} \implies C_{t+1} \text{ can be updated recursively.}$$

Theoretical Properties

Theorem 1: Ergodicity of the AM chain

Let π be the density of a target distribution supported on a bounded measurable subset $S \subset \mathbb{R}^d$, and assume that π is bounded from above.

Then the AM chain simulates properly the target distribution π : for any bounded and measurable function $f: S \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \frac{1}{n+1} (f(X_0) + f(X_1) + \cdots + f(X_n)) = \int_S f(x) \pi(dx) \quad \text{a.s.}$$

Corollary: Stabilization of the Covariance

Under the assumptions of Theorem 1, the covariance C_t almost surely stabilizes during the algorithm. In fact, as $t \rightarrow \infty$,

$$C_t \longrightarrow s_d \text{cov}(\pi) + s_d \varepsilon I_d \quad \text{a.s.}$$

where $\text{cov}(\pi)$ denotes the covariance of the target distribution π .

Theoretical Properties

- Let T be a transition probability on S and set

$$\delta(T) = \sup_{\mu_1 \neq \mu_2} \frac{\|\mu_1 T - \mu_2 T\|}{\|\mu_1 - \mu_2\|}$$

where the supremum is taken over distinct probability measures on $(S, \mathcal{B})$ and $\|\cdot\|$ denotes the total variation norm.

The condition $\delta(T^{k_0}) < 1$ for some $k_0 \geq 1$ is well known to be equivalent to the uniform ergodicity of the Markov chain having transition probability T .⁴

- Given a generalized transition probability K_n and a fixed $(n-1)$ -tuple

$$\tilde{y}_{n-2} = (y_0, y_1, \dots, y_{n-2}) \in S^{n-1}$$

Define

$$K_{n, \tilde{y}_{n-2}}(x, A) = K_n(y_0, y_1, \dots, y_{n-2}, x; A), \quad x \in S, A \in \mathcal{B}(S)$$

This is obtained by freezing the first $n-1$ variables.

⁴Nummelin 1984.

Theoretical Properties

Conditions of Theorem 2

Assume the sequence of generalized transition probabilities (K_n) satisfies:

- ① There are a fixed integer k_0 and a constant $\lambda \in (0, 1)$ such that

$$\delta\left((K_{n, \tilde{y}_{n-2}})^{k_0}\right) \leq \lambda < 1, \quad \forall \tilde{y}_{n-2} \in S^{n-1}, \quad n \geq 2.$$

- ② There are a fixed probability measure π on S and $c_0 > 0$ such that

$$\|\pi K_{n, \tilde{y}_{n-2}} - \pi\| \leq \frac{c_0}{n}, \quad \forall \tilde{y}_{n-2} \in S^{n-1}, \quad n \geq 2.$$

- ③ We have the following estimate for the operator norm:

$$\|K_{n, \tilde{y}_{n-2}} - K_{n+k, \tilde{y}_{n+k-2}}\| \leq c_1 \frac{k}{n},$$

where c_1 is a fixed positive constant and $\tilde{y}_{n+k-2}$ is a direct continuation of $\tilde{y}_{n-2}$.

Then, for any bounded and measurable $f: S \rightarrow \mathbb{R}$, $\frac{1}{n+1} \sum_{i=0}^n f(X_i) \xrightarrow{\text{a.s.}} \int_S f(x) \pi(dx)$.

AM: Practical Remarks

Remark 1: Early Stage

- Use any prior knowledge available (e.g., MLE, approximate Cov) to help choosing the initial state / C_0 .
- **Greedy Start:** update the proposal using only the accepted states during a short initial period.
- If the chain hasn't moved enough in the early stage, the proposal covariance can be shrunk by a constant factor.

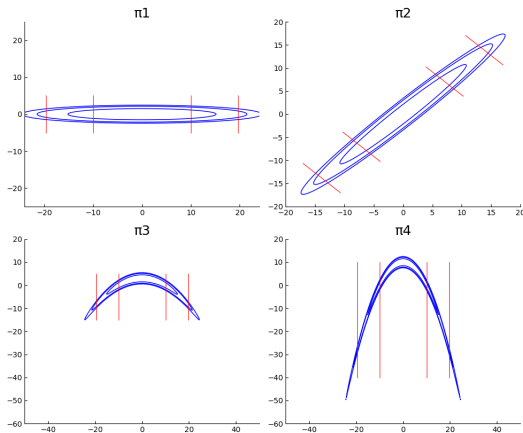
Remark 2: Update

- One may update the covariance only every n_0 -th step (again using the entire history).
- One may also determine the covariance by using only an increasing part of the near history.
- For example, one may determine C_n by using only the samples $X_{[n/2]}, X_{[n/2]+1}, \dots, X_n$.

Remark 3: Proposal

- Gaussian proposals yield a family of proposal distributions with a natural parametrization for size and orientation and which are easy to compute with.
- For example, one may use uniform distributions in a parallelepiped whose shape is guided by C_t .
- Tuning only according to the acceptance rate may be insufficient (see examples later).
- Some theoretical assumptions such as compact support and bounded target density may be weakened in practice.

Simulation 3: Four Target Distributions



The contours refer to 68.3%, 90% and 95% confidence regions in dimension 2. The lines separate the 1-dimensional confidence region between 68.3% and 95%.(最大方差方向)

Banana-shaped Distribution

Let f be the density of $\mathcal{N}(0, C_1)$ and define $f_b = f \circ \phi_b$, where

$$\phi_b(x) = (x_1, x_2 + bx_1^2 - 100b, x_3, \dots, x_n).$$

The function ϕ_b only changes the second coordinate, and the nonlinearity increases with b .

- π_1 : Uncorrelated Gaussian with covariance $C_1 = \text{diag}(100, 1, \dots, 1)$.
- π_2 : Rotated version of π_1 , whose main axis is rotated toward the direction $(1, \dots, 1)$.
- π_3 : Moderately non-linear banana-shaped distribution with $b = 0.03$.
- π_4 : Strongly non-linear banana-shaped distribution with $b = 0.1$.
- π_1 is restricted to a parallelepiped with corners at points $(\pm 35, \pm 3.5, \dots, \pm 3.5)$ which contains about 99.6% of the probability mass.

AP and SC Algorithms

Adaptive Proposal

- AP updates proposal covariance from the past H states every U steps.
- H and U should increase for larger dimensions d .
- In non-linear case compared with the linear case, we should also choose larger values for H and U .

Target	History H	Update frequency U
π_1	700	700
π_2	700	700
π_3	1600	1600
π_4	2000	2000

Single-Component Metropolis

Update only one component $y_j = x_j + \epsilon_j$, $\epsilon_j \sim N(0, \sigma_j^2)$ while keeping all other components fixed.

- For π_1 , the one-dimensional proposal distributions are chosen as

$$Q_i \sim N(0, C_1(i, i)), i = 1, \dots, d,$$

where $C_1 = \text{diag}(100, 1, \dots, 1)$.

- For π_2 , use the reasonable choice $Q_i \sim N(0, 1)$ for each coordinate.
- For π_3 and π_4 , the proposal standard deviations are chosen from C_1 . Thus, $(\sigma_1, \sigma_2, \dots, \sigma_d) = (10, 1, \dots, 1)$.

Simulation 3: Set up

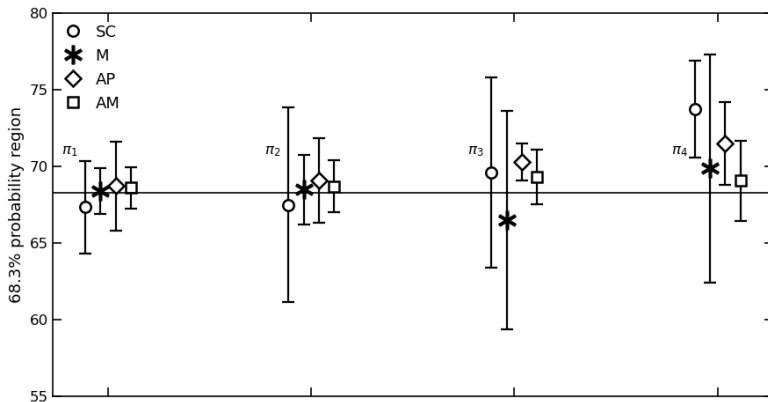
Initialization of the Proposal Covariance C_0

- For π_1 , the known covariance matrix is $C_1 = \text{diag}(100, 1, \dots, 1)$, thus the proposal covariance is chosen as $C_0 = \frac{2.38^2}{d} C_1$.
- For π_2 , the covariance matrix is the rotated version of C_1 : $C_2 = Q C_1 Q^T$, where Q is the rotation matrix. Therefore, $C_0 = \frac{2.38^2}{d} C_2$.
- For π_3 and π_4 , the initial proposal covariance is still chosen from C_1 : $C_0 = \frac{2.38^2}{d} C_1$.

Target	Number of iterations	Burn-in
π_1	20,000	10,000
π_2	20,000	10,000
π_3	40,000	20,000
π_4	80,000	40,000

Each experiment is repeated 100 times.

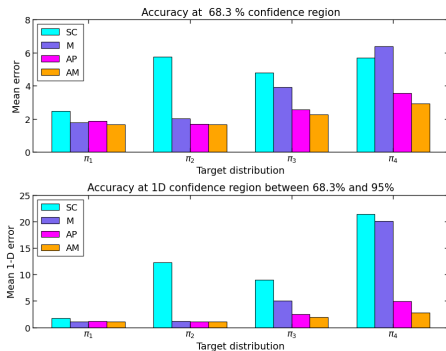
Simulation 3: Results



The symbols correspond to the mean frequency of hits in the 68.3% probability region of 100 simulations and the error bar around the symbol corresponds to the standard deviation of the hits. The true value (68.3%) is indicated by a horizontal line.

- Diagnostic: proportion of post-burn-in samples falling inside the true 68.3% probability region of the target distribution.
- If the algorithm samples correctly, this proportion should be close to 68.3%.
- For π_1 and π_2 , all four algorithms are reasonably close and SC has larger variation.
- For π_3 and π_4 , AM performs better and more stably on nonlinear targets.

Simulation 3: Results



- The top figure gives the mean of the absolute values of the errors for 68.3% confidence limit.
- The lower figure gives the mean of the absolute values of the errors for the 1-dimensional confidence region between 68.3% and 95%.

Target	SC	M	AP	AM
$err(\leq 68.3\%)$				
π_1	2.46	1.79	1.86	1.65
π_2	5.75	2.01	1.68	1.65
π_3	4.79	3.91	2.55	2.27
π_4	5.70	6.38	3.56	2.92
$1 - d \quad err(68.3\% - 95\%)$				
π_1	1.76	1.08	1.17	1.09
π_2	12.27	1.14	1.09	1.11
π_3	9.01	5.04	2.47	1.93
π_4	21.43	20.05	4.95	2.82

- AP and AM are clearly useful on nonlinear targets π_3 and π_4 .

Simulation 3: Appendix

- **mean**($\|E\|$) denotes the mean distance of the expectation values from the true value at origo calculated over 100 repetitions

$$\text{mean}(\|E\|) = \frac{1}{100} \sum_{j=1}^{100} \left(\sum_{i=1}^d (E_j^i)^2 \right)^{\frac{1}{2}}$$

where E_j is the mean vector of chain j .

- **std**($\|E\|$) denotes the standard deviation of the distances from the true value at origo.
- **err**($\leq 68.3\%$) denotes the mean absolute error of the percentages of sampled points that hit the confidence region inside the 68.3% level over repeated runs.
- **std**($\leq 68.3\%$) is the standard deviation of the percentages of the sampled points that hit the confidence region inside 68.3% level.
- **err**($\geq 99\%$) and **std**($\geq 99\%$) are defined analogously for the percentages of sampled points that hit the region outside the 99% confidence region.
- **1-d err**(68.3 – 95%) denotes the mean absolute error of the percentages of sampled points that hit the one-dimensional confidence region between the 68.3% and 95% limits. The 1-dimensional confidence region contains all those points whose projections to the direction of the main semi-axis of the target distribution belong to the corresponding 1-dimensional confidence interval. In the case of the non-linear target distributions π_3 and π_4 the main semi-axis is $[1, 0, \dots]$.
- **1-d std**(68.3 – 95%) is the standard deviation of the percentages of the sampled points that hit the 1-dimensional confidence region between 68.3% and 95%.
- **1-d err**(95 – 99%) and **1-d std**(95 – 99%) values are the corresponding ones between 95% and 99% limits.

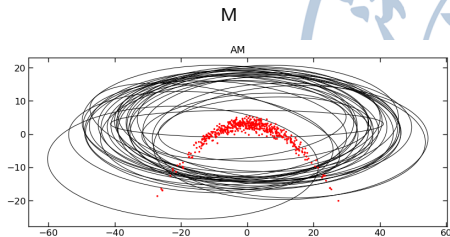
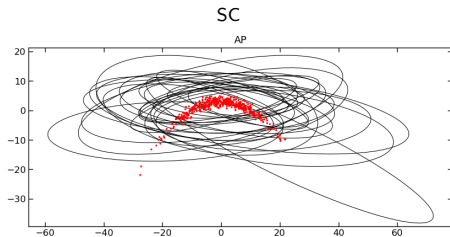
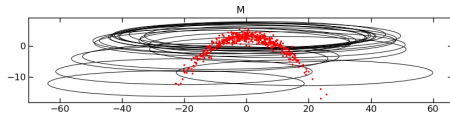
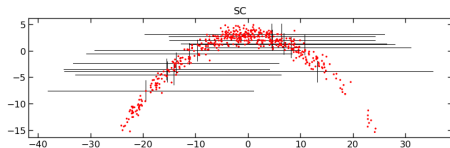
Simulation 3: Appendix

π_1					π_2				
Metric	SC	M	AP	AM	Metric	SC	M	AP	AM
mean($\ E\ $)	0.78	0.43	0.48	0.43	mean($\ E\ $)	5.86	0.41	0.43	0.46
std($\ E\ $)	0.50	0.29	0.31	0.24	std($\ E\ $)	4.20	0.24	0.29	0.28
err($\leq 68.3\%$)	2.46	1.79	1.86	1.65	err($\leq 68.3\%$)	5.75	2.01	1.68	1.65
std($\leq 68.3\%$)	2.98	2.15	2.32	1.96	std($\leq 68.3\%$)	7.73	2.41	2.11	2.04
err($\geq 99\%$)	0.38	0.30	0.31	0.31	err($\geq 99\%$)	0.56	0.36	0.32	0.32
std($\geq 99\%$)	0.38	0.33	0.32	0.28	std($\geq 99\%$)	0.77	0.35	0.33	0.34
1-d err(68.3-95%)	1.76	1.08	1.17	1.09	1-d err(68.3-95%)	12.27	1.14	1.09	1.11
1-d std(68.3-95%)	2.18	1.27	1.40	1.33	1-d std(68.3-95%)	15.25	1.38	1.37	1.41
1-d err(95-99%)	0.88	0.51	0.47	0.39	1-d err(95-99%)	5.46	0.50	0.42	0.40
1-d std(95-99%)	1.09	0.61	0.58	0.52	1-d std(95-99%)	8.20	0.62	0.54	0.52
acc. rate (%)	70.48	26.72	28.25	33.43	acc. rate (%)	72.13	26.69	28.28	33.64

π_3 ($b=0.03$)					π_4 ($b=0.1$)				
Metric	SC	M	AP	AM	Metric	SC	M	AP	AM
mean($\ E\ $)	3.21	2.19	1.41	1.04	mean($\ E\ $)	6.66	7.49	3.83	2.10
std($\ E\ $)	3.95	2.55	0.90	0.59	std($\ E\ $)	4.34	7.29	0.96	1.06
err($\leq 68.3\%$)	4.79	3.91	2.55	2.27	err($\leq 68.3\%$)	5.70	6.38	3.56	2.92
std($\leq 68.3\%$)	7.62	6.01	2.91	2.32	std($\leq 68.3\%$)	6.53	9.01	2.50	3.12
err($\geq 99\%$)	0.59	0.52	0.39	0.37	err($\geq 99\%$)	0.45	0.57	0.48	0.44
std($\geq 99\%$)	1.68	0.65	0.34	0.34	std($\geq 99\%$)	0.28	0.92	0.31	0.35
1-d err(68.3-95%)	9.01	5.04	2.47	1.93	1-d err(68.3-95%)	21.43	20.05	4.95	2.82
1-d std(68.3-95%)	11.23	6.28	3.12	2.40	1-d std(68.3-95%)	26.91	25.93	4.02	3.46
1-d err(95-99%)	4.68	3.68	1.86	1.32	1-d err(95-99%)	4.66	6.60	2.91	1.99
1-d std(95-99%)	7.04	5.09	2.28	1.56	1-d std(95-99%)	5.98	15.01	1.54	2.06
acc. rate (%)	66.05	15.55	15.08	17.47	acc. rate (%)	63.90	8.05	9.65	8.45



Simulation 3: Proposal Ellipses



An example of the simulated chain when the target function was 2-dimensional banana-shaped distribution π_3 . The ellipsoids (or line segments) refer to 95% confidence regions of different proposal distributions.

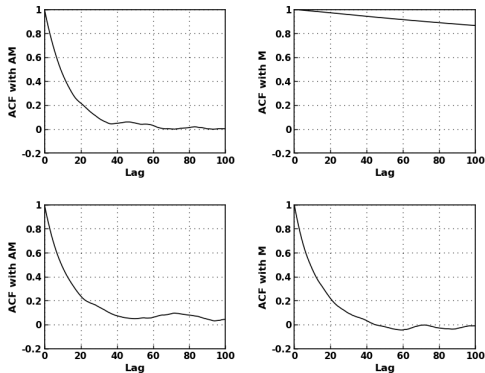
Why the Ellipses Matter

- **SC**: Only one coordinate is updated at each proposal step. The proposal interval in the first coordinate direction is much wider than that in the second coordinate direction. Although this algorithm can move efficiently along each coordinate direction separately, it cannot propose moves along the local tangent direction of the banana-shaped target.
- **M**: The proposal covariance is fixed throughout the simulation. Hence, the shape and orientation of the confidence ellipses do not change during the run.
- **AP**: proposal ellipses can change orientation and size using the most recent H samples. This is useful for local structure.
- **AM**: proposal covariance comes from the full history. It is more stable and learns the overall covariance structure, but the curvature may be smoothed out.

Connection to the Intro Experiment

Even when scale is reasonable, the proposal covariance structure determines movement efficiency in different directions.

Simulation 3: Acceptance Rate Is Not Enough



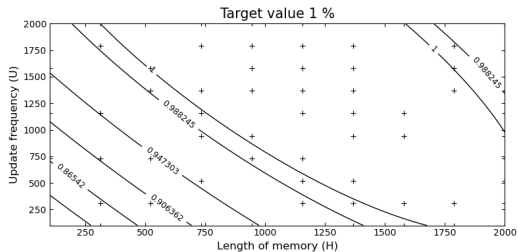
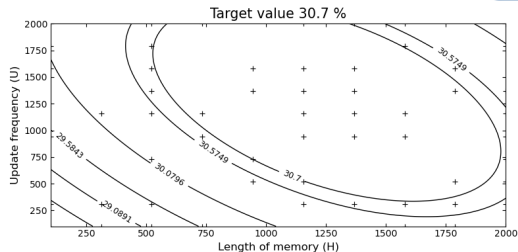
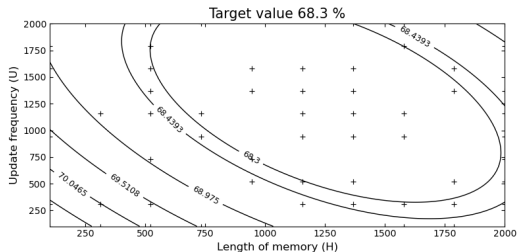
Comparison of the performance of the AM algorithm (left column) and the Metropolis algorithm with a fairly optimal acceptance rate (right column). The target distribution

was π_1 .

- Acceptance rate: AM 33% and M 25%.
- Top line: autocorrelation in the direction of the largest eigenvalue of target's covariance matrix.
- Bottom row: autocorrelation in an orthogonal direction.
- Tuning the proposal distribution according to the acceptance rate only may lead to difficulties.
- **Another example:** for highly multimodal target distributions, a chain may have a reasonable local acceptance rate while still moving very slowly between different modes.^a

^aGelman, Roberts, and Gilks 1996.

Simulation 4: Sensitivity with respect to the tuning parameters of AP



- Sensitivity of AP on memory (H) and update frequency (U) in 8-dimensional Gaussian target (π_1).
- A quadratic regression model was found to adequately explain the response variables tested: the ratio of visits inside the 68.3% region, between the 68.3% and 99% regions, and outside the 99% region.
- The response variables were taken as averages over 100 repetitions.

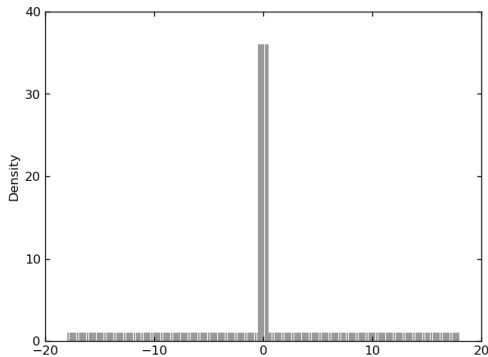
Simulation 5: Setup

An Example shows the Bias of AP

Consider a density f defined on the rectangle $R = [-18, 18] \times [-3, 3] \subset \mathbb{R}^2$ containing the middle strip $S = [-0.5, 0.5] \times [-3, 3]$.

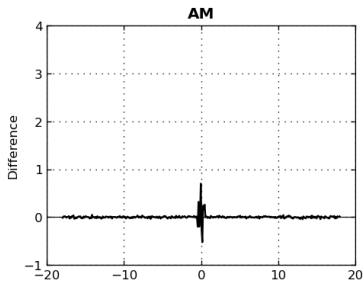
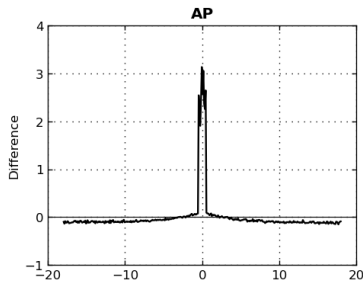
$$f(x) = \begin{cases} 36 & \text{if } x \in S \\ 1 & \text{if } x \in R \setminus S \end{cases}$$

- $\pi(S) : \pi(R \setminus S) = 36 : 35$ means about half of the mass is concentrated on S .
- An ergodic MCMC algorithm should stay for about the **same** amount of time on S and on R .
- S and R are thin rectangles with **opposite** orientations.
- AP algorithm has to regularly turn the direction of the proposal distribution.
- This causes notable bias in the simulation on S .



The (unscaled) one-dimensional projection of true target distribution

Simulation 5: Considerable Error of AP in Specific Cases



The difference between the sampled distributions and the real target distribution.

$$(\text{difference} = \text{sampled density} - \text{true density})$$

- The curve represents the mean values of 100 runs with 100 000 states.
- For AP, the relative error in the simulation on S is about 10%. There is also a slight error in the simulation near the far ends of the rectangle R .
- This phenomenon underlines the importance of calculating the covariance from an increasing segment of the history, as is done in the AM algorithm.
- Outside $\rightarrow$ Central: $\alpha = 1$. Central $\rightarrow$ Outside $\alpha = 1/36$.
- Once the chain remains in the central strip for some time, the proposal of AP becomes a vertically elongated ellipse, making it hard to move out.

Simulation 6: Instability of AM for a Multivariate Student-t Target

Target

A bivariate Student distribution with $n = 1$ degrees of freedom and the following location and pseudo-covariance matrix

$$\mu = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 0.2 & 0.1 \\ 0.1 & 0.8 \end{bmatrix}$$

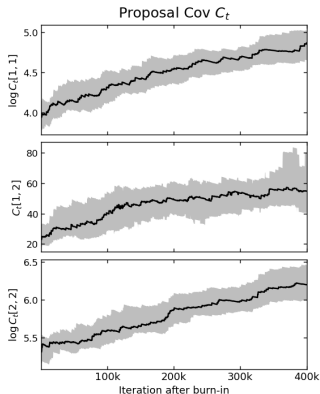
respectively.

That is, the target density $\pi(x) \propto [1 + (x - \mu)^T \Sigma^{-1} (x - \mu)]^{-3/2}$.

- π has no second moments (no $\text{Cov}(\pi)$).
- Thereby the empirical covariance estimate used by AM is deemed to be unstable in this example.⁵

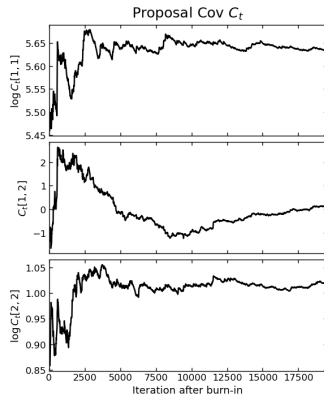
⁵Vihola 2011.

Simulation 6: Results

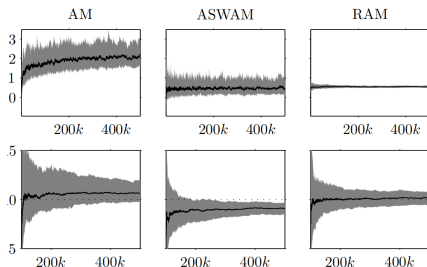
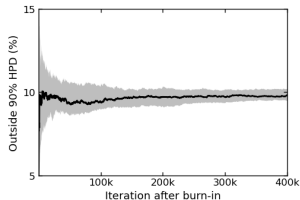


Bivariate Student Distribution

- The grey area indicates the interval between the 37.5% and the 62.5% percentiles, and the black line shows the median.
- For Gaussian targets, the proposal covariance stabilizes, whereas for Student-t targets without finite second moments, it keeps growing.

Normal Distribution π_1

Simulation 6: Partial Reproduction but No HPD Bias



the proportion of X_n outside the 90% highest probability
density of the target

算法 Robust Adaptive Metropolis (RAM)

- 1: Proposal density q is spherically symmetric, $\{\eta_n\}$ is a step size sequence decaying to zero and α_* stand for the target mean acceptance probability of the algorithm.
- 2: Draw $U_n \sim q$ and set $Y_n = X_{n-1} + S_{n-1} U_n$.
- 3: Compute $\alpha_n = \min \left\{ 1, \frac{\pi(Y_n)}{\pi(X_{n-1})} \right\}$.
- 4: With probability α_n set $X_n = Y_n$; otherwise $X_n = X_{n-1}$. Update S_n satisfying

$$S_n S_n^T = S_{n-1} \left(I + \eta_n (\alpha_n - \alpha_*) \frac{U_n U_n^T}{\|U_n\|^2} \right) S_{n-1}^T.$$

Conclusion

- ① **Scale matters.** A proposal that is too small accepts often but moves slowly; a proposal that is too large is rejected often. Gelman *et al.*⁶ gives $2.38/\sqrt{d}$ and the high-dimensional 0.234 acceptance-rate rule.
- ② **Shape also matters.** Even with the same overall scale, a better covariance structure can reduce autocorrelation and improve mixing.
- ③ **Adaptive covariance is a natural solution.** AP learns local covariance from recent samples but may introduce bias. AM uses the full history, preserving adaptive covariance learning while obtaining correct ergodic properties.

⁶Gelman, Roberts, and Gilks 1996.

Recent Directions: Beyond Global Covariance Adaptation

Year	Work	Main idea
2006	Andrieu & Moulines <i>Ergodicity of adaptive MCMC</i>	Provides general convergence tools for adaptive algorithms, including conditions under which ergodic averages and CLT-type results can hold.
2007	Roberts & Rosenthal <i>Coupling and ergodicity</i>	Introduces the widely used <i>diminishing adaptation</i> and <i>containment</i> framework for proving validity of adaptive MCMC.
2010	Saksman & Vihola <i>AM on unbounded domains</i>	Extends ergodicity theory for Adaptive Metropolis beyond bounded support, under tail and regular-contour conditions.
2012	Vihola <i>Robust Adaptive Metropolis</i>	Learns proposal shape while coercing the acceptance rate toward a target. More robust when empirical covariance is unstable or moments may not exist.
2013	Sejdinovic et al. <i>Kernel Adaptive MH</i>	Uses RKHS / kernel ideas to construct proposals with local covariance, targeting nonlinear supports where one global covariance is inadequate.
2019–2022	Titsias & Dellaportas; Yang–Roberts–Rosenthal; Schmon & Gagnon	Connects adaptive proposal learning with gradient-based objectives, general high-dimensional targets, and Bayesian large-sample asymptotics.
2021–	Normalizing-flow assisted MCMC	Uses learned transport maps or generative proposals to capture nonlinear, multimodal, or nonlocal geometry beyond Gaussian covariance adaptation.

References

-  Gelman, A., Roberts, G. O., and Gilks, W. R. (1996).
Efficient Metropolis jumping rules
-  Roberts, G. O., Gelman, A., and Gilks, W. R. (1997).
Weak convergence and optimal scaling of random walk Metropolis algorithms
-  Haario, H., Saksman, E., and Tamminen, J. (1999).
Adaptive proposal distribution for random walk Metropolis algorithm
-  Haario, H., Saksman, E., and Tamminen, J. (2001).
An adaptive Metropolis algorithm

